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## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Localizer

TA: 9 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | On                 |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | Off                |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | On                 |
| Load Images to Graphic Segments               | On                 |
| Graphic segment                               | All Segments       |
| Inline Movie                                  | Off                |

**Routine**

|                     |                    |
|---------------------|--------------------|
| Slice Group         | 1                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 2                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 3                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Phase Oversampling  | 0 %                |
| FoV Read            | 300 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 5.0 mm             |
| TR                  | 40.0 ms            |
| TE                  | 3.00 ms            |
| Averages            | 1                  |
| Concatenations      | 1                  |
| AutoAlign           | ---                |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| TR                 | 40.0 ms   |
| TE                 | 3.00 ms   |
| MTC                | Off       |
| Magn. Preparation  | None      |
| Flip Angle         | 15 deg    |
| Fat-Water Contrast | Standard  |
| Dark Blood         | Off       |
| Contrasts          | 1         |
| SWI                | Off       |
| Reconstruction     | Magnitude |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Measurements    | 1        |
| Multiple Series | Off      |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 300 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 5.0 mm  |
| Base Resolution  | 256     |
| Phase Resolution | 75 %    |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                         |         |
|-------------------------|---------|
| Acceleration mode       | None    |
| Advanced Reconstruction | Off     |
| Phase Partial Fourier   | Off     |
| Asymmetric Echo         | Allowed |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | On      |
| Distortion Correction | 2D      |
| Normalize             | Prescan |
| Image Filter          | Off     |

**Geometry - Common**

|                     |                    |
|---------------------|--------------------|
| Slice Group         | 1                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 2                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 3                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Phase Oversampling  | 0 %                |
| FoV Read            | 300 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 5.0 mm             |
| TR                  | 40.0 ms            |
| Multi-Slice Mode    | Interleaved        |
| Series              | Interleaved        |
| Concatenations      | 1                  |

**Geometry - AutoAlign**

|                     |                    |
|---------------------|--------------------|
| Slice Group         | 1                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 2                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 3                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Transversal        |

**Geometry - AutoAlign**

|                     |                 |
|---------------------|-----------------|
| Phase Encoding Dir. | A >> P          |
| AutoAlign           | ---             |
| Initial Position    | L0.0 A20.0 H0.0 |
| L                   | 0.0 mm          |
| A                   | 20.0 mm         |
| H                   | 0.0 mm          |
| Initial Orientation | Transversal     |
| Initial Rotation    | 0.00 deg        |

**Geometry - Saturation**

|                    |          |
|--------------------|----------|
| Saturation Mode    | Standard |
| Special Saturation | None     |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Off               |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Tune up  |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|              |            |
|--------------|------------|
| B1 Shim      | TrueForm   |
| Excitation   | Slice-sel. |
| LR Balancing | Off        |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |         |
|-----------------|---------|
| 1st Signal/Mode | None    |
| TR              | 40.0 ms |
| Segments        | 1       |

**Physio - Signal**

|                |   |
|----------------|---|
| Concatenations | 1 |
|----------------|---|

**Physio - Cardiac**

|                    |          |
|--------------------|----------|
| Tagging            | None     |
| Fat-Water Contrast | Standard |
| Magn. Preparation  | None     |
| Dark Blood         | Off      |
| FoV Read           | 300 mm   |
| FoV Phase          | 100.0 %  |
| Phase Resolution   | 75 %     |
| Dynamic Mode       | Standard |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 1   |

**Inline - Liver**

|                      |     |
|----------------------|-----|
| Liver Registration   | Off |
| Save Original Images | On  |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |         |
|----------------------|---------|
| Magn. Preparation    | None    |
| Save Original Images | On      |
| Contrasts            | 1       |
| TE                   | 3.00 ms |
| TR                   | 40.0 ms |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Soft Tissue**

|              |     |
|--------------|-----|
| Wash-in      | Off |
| Wash-out     | Off |
| TTP          | Off |
| PEI          | Off |
| MIP Time     | Off |
| Measurements | 1   |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Inline - MapIt**

|                      |         |
|----------------------|---------|
| MapIt                | None    |
| Flip Angle           | 15 deg  |
| Measurements         | 1       |
| Contrasts            | 1       |
| TE                   | 3.00 ms |
| TR                   | 40.0 ms |
| Save Original Images | On      |

**Sequence - Part 1**

|                   |            |
|-------------------|------------|
| Sequence Name     | fl         |
| Dimension         | 2D         |
| Excitation        | Slice-sel. |
| RF Pulse Type     | Normal     |
| Gradient Mode     | Fast       |
| Flow Compensation | None       |
| Bandwidth         | 260 Hz/Px  |
| Asymmetric Echo   | Allowed    |
| Segments          | 1          |

**Sequence - Part 2**

|                          |     |
|--------------------------|-----|
| Introduction             | On  |
| RF Spoiling              | On  |
| Acoustic noise reduction | Off |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
| Allowed Delay | 0 s |

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\AAHScout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm<sup>3</sup> Acc:: 3 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | Off                |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | On                 |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |      |
|------------------------|------|
| Acceleration Factor 3D | 1    |
| Phase Partial Fourier  | 6/8  |
| Slice Partial Fourier  | 6/8  |
| Asymmetric Echo        | Weak |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | Off     |
| Distortion Correction | 3D      |
| Normalize             | Prescan |
| Noise Masking         | Off     |
| Image Filter          | Off     |

**Routine**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 128                |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FoV Read            | 260 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 1.6 mm             |
| TR                  | 3.2 ms             |
| TE                  | 1.37 ms            |
| Averages            | 1                  |
| Concatenations      | 1                  |
| AutoAlign           | Head               |

**Geometry - Common**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 128                |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FoV Read            | 260 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 1.6 mm             |
| TR                  | 3.2 ms             |
| Multi-Slice Mode    | Sequential         |
| Series              | Ascending          |
| Concatenations      | 1                  |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| TR                 | 3.2 ms    |
| TE                 | 1.37 ms   |
| Flip Angle         | 8 deg     |
| Fat-Water Contrast | Standard  |
| Contrasts          | 1         |
| Reconstruction     | Magnitude |

**Geometry - AutoAlign**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| AutoAlign           | Head               |
| Initial Position    | Isocenter          |
| L                   | 0.0 mm             |
| P                   | 0.0 mm             |
| H                   | 0.0 mm             |
| Initial Orientation | Transversal        |
| Initial Rotation    | 0.00 deg           |

**Contrast - Dynamic**

|                |          |
|----------------|----------|
| Dynamic Mode   | Standard |
| Measurements   | 1        |
| Time to Center | 6.2 s    |

**Resolution - Common**

|                  |           |
|------------------|-----------|
| FoV Read         | 260 mm    |
| FoV Phase        | 100.0 %   |
| Slice Thickness  | 1.6 mm    |
| Base Resolution  | 160       |
| Phase Resolution | 100 %     |
| Slice Resolution | 69 %      |
| Trajectory       | Cartesian |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Adaptive Combine  |
| Matrix Optimization | Off               |
| Coil Focus          | Flat              |

**Resolution - Acceleration**

|                        |            |
|------------------------|------------|
| Acceleration mode      | GRAPPA     |
| Reference Scans        | Integrated |
| Acceleration Factor PE | 3          |
| Reference Lines PE     | 24         |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Tune up  |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Non-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 1   |

**Inline - Dynamic**

|                |          |
|----------------|----------|
| Dynamic Mode   | Standard |
| Flip Angle     | 8 deg    |
| Measurements   | 1        |
| Time to Center | 6.2 s    |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |         |
|----------------------|---------|
| Save Original Images | On      |
| Contrasts            | 1       |
| TE                   | 1.37 ms |
| TR                   | 3.2 ms  |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Inline - MapIt**

|                      |         |
|----------------------|---------|
| MapIt                | None    |
| Flip Angle           | 8 deg   |
| Measurements         | 1       |
| Contrasts            | 1       |
| TE                   | 1.37 ms |
| TR                   | 3.2 ms  |
| Save Original Images | On      |

**Sequence - Part 1**

|                 |           |
|-----------------|-----------|
| Sequence Name   | fl        |
| Dimension       | 3D        |
| Excitation      | Non-sel.  |
| RF Pulse Type   | Fast      |
| Gradient Mode   | Normal    |
| Bandwidth       | 540 Hz/Px |
| Asymmetric Echo | Weak      |

**Sequence - Part 2**

|                    |     |
|--------------------|-----|
| Introduction       | On  |
| RF Spoiling        | On  |
| Breast Application | Off |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Localizer\_aligned

TA: 21 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm<sup>3</sup> Acc.: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | Off                |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | On                 |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | On                 |
| Load Images to Graphic Segments               | On                 |
| Graphic segment                               | All Segments       |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 300 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 5.0 mm  |
| Base Resolution  | 256     |
| Phase Resolution | 75 %    |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                         |         |
|-------------------------|---------|
| Acceleration mode       | None    |
| Advanced Reconstruction | Off     |
| Phase Partial Fourier   | Off     |
| Asymmetric Echo         | Allowed |

**Routine**

|                     |              |
|---------------------|--------------|
| Slice Group         | 1            |
| Slices              | 1            |
| Distance Factor     | 20 %         |
| Position            | Isocenter    |
| Orientation         | Transversal  |
| Phase Encoding Dir. | A >> P       |
| Slice Group         | 2            |
| Slices              | 1            |
| Distance Factor     | 20 %         |
| Position            | Isocenter    |
| Orientation         | Transversal  |
| Phase Encoding Dir. | A >> P       |
| Slice Group         | 3            |
| Slices              | 1            |
| Distance Factor     | 20 %         |
| Position            | Isocenter    |
| Orientation         | Transversal  |
| Phase Encoding Dir. | A >> P       |
| Phase Oversampling  | 0 %          |
| FoV Read            | 300 mm       |
| FoV Phase           | 100.0 %      |
| Slice Thickness     | 5.0 mm       |
| TR                  | 104.0 ms     |
| TE                  | 3.00 ms      |
| Averages            | 1            |
| Concatenations      | 1            |
| AutoAlign           | Head > Brain |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | On      |
| Distortion Correction | 2D      |
| Normalize             | Prescan |
| Image Filter          | Off     |

**Geometry - Common**

|                     |             |
|---------------------|-------------|
| Slice Group         | 1           |
| Slices              | 1           |
| Distance Factor     | 20 %        |
| Position            | Isocenter   |
| Orientation         | Transversal |
| Phase Encoding Dir. | A >> P      |
| Slice Group         | 2           |
| Slices              | 1           |
| Distance Factor     | 20 %        |
| Position            | Isocenter   |
| Orientation         | Transversal |
| Phase Encoding Dir. | A >> P      |
| Slice Group         | 3           |
| Slices              | 1           |
| Distance Factor     | 20 %        |
| Position            | Isocenter   |
| Orientation         | Transversal |
| Phase Encoding Dir. | A >> P      |
| Phase Oversampling  | 0 %         |
| FoV Read            | 300 mm      |
| FoV Phase           | 100.0 %     |
| Slice Thickness     | 5.0 mm      |
| TR                  | 104.0 ms    |
| Multi-Slice Mode    | Interleaved |
| Series              | Interleaved |
| Concatenations      | 1           |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| TR                 | 104.0 ms  |
| TE                 | 3.00 ms   |
| MTC                | Off       |
| Magn. Preparation  | None      |
| Flip Angle         | 15 deg    |
| Fat-Water Contrast | Standard  |
| Dark Blood         | Off       |
| Contrasts          | 1         |
| SWI                | Off       |
| Reconstruction     | Magnitude |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Measurements    | 1        |
| Multiple Series | Off      |

**Geometry - AutoAlign**

|                     |             |
|---------------------|-------------|
| Slice Group         | 1           |
| Position            | Isocenter   |
| Orientation         | Transversal |
| Phase Encoding Dir. | A >> P      |
| Slice Group         | 2           |
| Position            | Isocenter   |
| Orientation         | Transversal |
| Phase Encoding Dir. | A >> P      |
| Slice Group         | 3           |
| Position            | Isocenter   |
| Orientation         | Transversal |

**Geometry - AutoAlign**

|                     |              |
|---------------------|--------------|
| Phase Encoding Dir. | A >> P       |
| AutoAlign           | Head > Brain |
| Initial Position    | Isocenter    |
| L                   | 0.0 mm       |
| P                   | 0.0 mm       |
| H                   | 0.0 mm       |
| Initial Orientation | Transversal  |
| Initial Rotation    | 0.00 deg     |

**Geometry - Saturation**

|                    |          |
|--------------------|----------|
| Saturation Mode    | Standard |
| Special Saturation | None     |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Off               |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Tune up  |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|              |            |
|--------------|------------|
| B1 Shim      | TrueForm   |
| Excitation   | Slice-sel. |
| LR Balancing | Off        |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |          |
|-----------------|----------|
| 1st Signal/Mode | None     |
| TR              | 104.0 ms |
| Segments        | 1        |

**Physio - Signal**

|                |   |
|----------------|---|
| Concatenations | 1 |
|----------------|---|

**Physio - Cardiac**

|                    |          |
|--------------------|----------|
| Tagging            | None     |
| Fat-Water Contrast | Standard |
| Magn. Preparation  | None     |
| Dark Blood         | Off      |
| FoV Read           | 300 mm   |
| FoV Phase          | 100.0 %  |
| Phase Resolution   | 75 %     |
| Dynamic Mode       | Standard |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 1   |

**Inline - Liver**

|                      |     |
|----------------------|-----|
| Liver Registration   | Off |
| Save Original Images | On  |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |          |
|----------------------|----------|
| Magn. Preparation    | None     |
| Save Original Images | On       |
| Contrasts            | 1        |
| TE                   | 3.00 ms  |
| TR                   | 104.0 ms |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Soft Tissue**

|              |     |
|--------------|-----|
| Wash-in      | Off |
| Wash-out     | Off |
| TTP          | Off |
| PEI          | Off |
| MIP Time     | Off |
| Measurements | 1   |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Inline - MapIt**

|                      |          |
|----------------------|----------|
| MapIt                | None     |
| Flip Angle           | 15 deg   |
| Measurements         | 1        |
| Contrasts            | 1        |
| TE                   | 3.00 ms  |
| TR                   | 104.0 ms |
| Save Original Images | On       |



**Sequence - Part 1**

|                   |            |
|-------------------|------------|
| Sequence Name     | fl         |
| Dimension         | 2D         |
| Excitation        | Slice-sel. |
| RF Pulse Type     | Normal     |
| Gradient Mode     | Fast       |
| Flow Compensation | None       |
| Bandwidth         | 260 Hz/Px  |
| Asymmetric Echo   | Allowed    |
| Segments          | 1          |

**Sequence - Part 2**

|                          |     |
|--------------------------|-----|
| Introduction             | On  |
| RF Spoiling              | On  |
| Acoustic noise reduction | Off |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
| Allowed Delay | 0 s |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\T1w\_setter\_64ch

TA: 1 sec Coil Selection: Manual Voxel Size: 4.0×4.0×8.0 mm³ Rel. SNR: 1.00

**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 50 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 32                 |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FoV Read            | 256 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 8.0 mm             |
| TR                  | 19.0 ms            |
| TE                  | 8.50 ms            |
| Averages            | 1                  |
| Concatenations      | 1                  |
| AutoAlign           | ---                |
| Coil Elements       | BC                 |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| TR                 | 19.0 ms   |
| TE                 | 8.50 ms   |
| MTC                | Off       |
| Flip Angle         | 2 deg     |
| Fat-Water Contrast | Standard  |
| Reconstruction     | Magnitude |

**Contrast - Dynamic**

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 256 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 8.0 mm  |
| Base Resolution  | 64      |
| Phase Resolution | 100 %   |
| Slice Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |     |
|-----------------------|-----|
| Phase Partial Fourier | Off |
| Slice Partial Fourier | 6/8 |

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Hamming               | Off |
| Distortion Correction | Off |
| Normalize             | Off |

**Geometry - Common**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 50 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 32                 |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FoV Read            | 256 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 8.0 mm             |
| TR                  | 19.0 ms            |
| Multi-Slice Mode    | Interleaved        |
| Series              | Interleaved        |
| Concatenations      | 1                  |

**Geometry - AutoAlign**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| AutoAlign           | ---                |
| Initial Position    | L0.0 A20.0 H0.0    |
| L                   | 0.0 mm             |
| A                   | 20.0 mm            |
| H                   | 0.0 mm             |
| Initial Orientation | Sagittal           |
| Initial Rotation    | 0.00 deg           |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                |
|---------------------|----------------|
| Coil Selection      | Manual         |
| MSMA                | S - C - T      |
| Sagittal            | R >> L         |
| Coronal             | A >> P         |
| Transversal         | F >> H         |
| Coil Combination    | Sum of Squares |
| Matrix Optimization | Off            |

**System - Adjustments**

|                      |          |
|----------------------|----------|
| Adjustment Strategy  | Standard |
| B0 Shim              | Tune up  |
| B1 Shim              | TrueForm |
| Adjustment Tolerance | Auto     |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| Adjust with Body Coil | On    |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Non-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Correction Factor   | 1.00           |
| Image Scaling       | 8.000          |

**Physio - Signal**

|                 |         |
|-----------------|---------|
| 1st Signal/Mode | None    |
| TR              | 19.0 ms |
| Concatenations  | 1       |

**Sequence - Part 1**

|                   |            |
|-------------------|------------|
| Sequence Name     | ABCD       |
| Dimension         | 3D         |
| Excitation        | Non-sel.   |
| RF Pulse Type     | Normal     |
| Gradient Mode     | Fast       |
| Bandwidth         | 4882 Hz/Px |
| Echo Spacing      | 0.49 ms    |
| Free Echo Spacing | Off        |
| EPI Factor        | 32         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | On  |

**Sequence - Special**

|                   |        |
|-------------------|--------|
| Protocol filename | MPRAGE |
|-------------------|--------|

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\T1w\_MPR\_vNav\_4e\_64c  
h

TA: 7:04 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc.: 2 Rel. SNR: 1.00

### Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | On                 |
| Graphic segment                               | 2nd Segment        |
| Inline Movie                                  | Off                |

### Routine

|                     |                     |
|---------------------|---------------------|
| Slab Group          | 1                   |
| Slabs               | 1                   |
| Distance Factor     | 50 %                |
| Position            | L0.0 A20.0 F30.0 mm |
| Orientation         | Sagittal            |
| Phase Encoding Dir. | A >> P              |
| Slices per Slab     | 208                 |
| Phase Oversampling  | 0 %                 |
| Slice Oversampling  | 7.7 %               |
| FoV Read            | 256 mm              |
| FoV Phase           | 93.8 %              |
| Slice Thickness     | 0.8 mm              |
| TR                  | 2500.0 ms           |
| TE 1                | 1.81 ms             |
| TE 2                | 3.60 ms             |
| TE 3                | 5.39 ms             |
| TE 4                | 7.27 ms             |
| Averages            | 1                   |
| Concatenations      | 1                   |
| AutoAlign           | ---                 |

### Contrast - Common

|                    |                       |
|--------------------|-----------------------|
| TR                 | 2500.0 ms             |
| TE 1               | 1.81 ms               |
| TE 2               | 3.60 ms               |
| TE 3               | 5.39 ms               |
| TE 4               | 7.27 ms               |
| Magn. Preparation  | Non-sel. IR           |
| TI                 | 1000 ms               |
| Flip Angle         | 8 deg                 |
| Fat-Water Contrast | Fast Water Excitation |
| Dark Blood         | Off                   |
| Contrasts          | 4                     |
| Reconstruction     | Magnitude             |

### Contrast - Dynamic

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |
| Reordering      | Linear           |

### Resolution - Common

|           |        |
|-----------|--------|
| FoV Read  | 256 mm |
| FoV Phase | 93.8 % |

### Resolution - Common

|                  |        |
|------------------|--------|
| Slice Thickness  | 0.8 mm |
| Base Resolution  | 320    |
| Phase Resolution | 100 %  |
| Slice Resolution | 100 %  |
| Interpolation    | Off    |

### Resolution - Acceleration

|                        |            |
|------------------------|------------|
| Acceleration mode      | GRAPPA     |
| Reference Scans        | Integrated |
| Acceleration Factor PE | 2          |
| Reference Lines PE     | 32         |
| Acceleration Factor 3D | 1          |
| Phase Partial Fourier  | Off        |
| Slice Partial Fourier  | 6/8        |
| Asymmetric Echo        | Off        |
| Elliptical Scanning    | Off        |

### Resolution - Filter

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | Off     |
| Distortion Correction | Off     |
| Normalize             | Prescan |
| Image Filter          | Off     |

### Geometry - Common

|                     |                     |
|---------------------|---------------------|
| Slab Group          | 1                   |
| Slabs               | 1                   |
| Distance Factor     | 50 %                |
| Position            | L0.0 A20.0 F30.0 mm |
| Orientation         | Sagittal            |
| Phase Encoding Dir. | A >> P              |
| Slices per Slab     | 208                 |
| Phase Oversampling  | 0 %                 |
| Slice Oversampling  | 7.7 %               |
| FoV Read            | 256 mm              |
| FoV Phase           | 93.8 %              |
| Slice Thickness     | 0.8 mm              |
| TR                  | 2500.0 ms           |
| Multi-Slice Mode    | Single Shot         |
| Series              | Interleaved         |
| Concatenations      | 1                   |

### Geometry - AutoAlign

|                     |                     |
|---------------------|---------------------|
| Slab Group          | 1                   |
| Position            | L0.0 A20.0 F30.0 mm |
| Orientation         | Sagittal            |
| Phase Encoding Dir. | A >> P              |
| AutoAlign           | ---                 |
| Initial Position    | L0.0 A20.0 F30.0    |
| R                   | 0.0 mm              |
| A                   | 20.0 mm             |
| F                   | 30.0 mm             |
| Initial Orientation | Sagittal            |
| Initial Rotation    | 0.00 deg            |

### Geometry - Tim Planning Suite

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |

**Geometry - Tim Planning Suite**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Adaptive Combine  |
| Matrix Optimization | Off               |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                     |
|-------------|---------------------|
| Position    | L0.0 A20.0 F30.0 mm |
| Orientation | Sagittal            |
| Rotation    | 0.00 deg            |
| A >> P      | 240 mm              |
| F >> H      | 256 mm              |
| R >> L      | 167 mm              |
| Reset       | Off                 |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Non-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 5.000          |

**Physio - Signal**

|                 |           |
|-----------------|-----------|
| 1st Signal/Mode | None      |
| TR              | 2500.0 ms |
| Concatenations  | 1         |

**Physio - Cardiac**

|                    |                       |
|--------------------|-----------------------|
| Fat-Water Contrast | Fast Water Excitation |
| Magn. Preparation  | Non-sel. IR           |
| TI                 | 1000 ms               |
| Dark Blood         | Off                   |
| FoV Read           | 256 mm                |
| FoV Phase          | 93.8 %                |
| Phase Resolution   | 100 %                 |
| Dynamic Mode       | Standard              |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |             |
|----------------------|-------------|
| Magn. Preparation    | Non-sel. IR |
| Save Original Images | On          |
| Contrasts            | 4           |
| TE 1                 | 1.81 ms     |
| TE 2                 | 3.60 ms     |
| TE 3                 | 5.39 ms     |
| TE 4                 | 7.27 ms     |
| TR                   | 2500.0 ms   |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Inline - MapIt**

|                      |           |
|----------------------|-----------|
| MapIt                | None      |
| Flip Angle           | 8 deg     |
| Measurements         | 1         |
| Contrasts            | 4         |
| TE 1                 | 1.81 ms   |
| TE 2                 | 3.60 ms   |
| TE 3                 | 5.39 ms   |
| TE 4                 | 7.27 ms   |
| TR                   | 2500.0 ms |
| Save Original Images | On        |

**Sequence - Part 1**

|                     |           |
|---------------------|-----------|
| Sequence Name       | tfl_me    |
| Dimension           | 3D        |
| Excitation          | Non-sel.  |
| RF Pulse Type       | Fast      |
| Gradient Mode       | Fast      |
| Flow Compensation 1 | None      |
| Flow Compensation 2 | None      |
| Flow Compensation 3 | None      |
| Flow Compensation 4 | None      |
| Reordering          | Linear    |
| Bandwidth 1         | 740 Hz/Px |
| Bandwidth 2         | 740 Hz/Px |
| Bandwidth 3         | 740 Hz/Px |
| Bandwidth 4         | 650 Hz/Px |
| Echo Spacing        | 11.40 ms  |
| Asymmetric Echo     | Off       |
| Turbo Factor        | 168       |

**Sequence - Part 2**

|                         |     |
|-------------------------|-----|
| Introduction            | On  |
| RF Spoiling             | On  |
| Incr. Gradient Spoiling | Off |

**Sequence - Special**

|                  |                |
|------------------|----------------|
| Readout polarity | Positive       |
| Nav. location    | None           |
| Apply moco to    | parent and nav |
| Remeasure        | 0 TRs          |

**Sequence - Special**

|                   |                |
|-------------------|----------------|
| Reacq. threshold  | 0.50           |
| Feedback Delay    | 60 ms          |
| Moco ref. image   | Use Temp Ref   |
| K-space streaming | None           |
| ABCD navigator    | On             |
| Add. grad time    | 0.00 ms        |
| Apply freq to     | parent and nav |
| Averaging         | RMS            |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\SpinEchoFieldMap\_AP

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FoV Read                 | 208 mm            |
| FoV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8000.0 ms         |
| TE                       | 66.00 ms          |
| Multi-band accel. factor | 1                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 8000.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Measurements    | 1        |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Hamming               | Off |
| Distortion Correction | Off |
| Normalize             | Off |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FoV Read                 | 208 mm            |
| FoV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8000.0 ms         |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 1                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| CoilShim              | Off   |
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | Off   |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8000.0 ms |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |         |
|-------------------------|---------|
| GLM Statistics          | Off     |
| Ignore Meas. at Start   | 0       |
| Ignore After Transition | 0       |
| Model Transition States | Off     |
| Temp. Highpass Filter   | Off     |
| Threshold               | 4.00    |
| Paradigm Size           | 3       |
| Meas[1]                 | Active  |
| Meas[2]                 | Active  |
| Meas[3]                 | Ignore  |
| Motion Correction       | Off     |
| Spatial Filter          | Off     |
| Measurements            | 1       |
| Delay in TR             | 0.00 ms |
| Multiple Series         | Off     |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                 |   |
|-----------------|---|
| Min. prep scans | 0 |
|-----------------|---|

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| SENSE1 coil combine      | Off        |
| Invert RO/PE polarity    | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |



\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\SpinEchoFieldMap\_PA

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FoV Read                 | 208 mm            |
| FoV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8000.0 ms         |
| TE                       | 66.00 ms          |
| Multi-band accel. factor | 1                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 8000.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Measurements    | 1        |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Hamming               | Off |
| Distortion Correction | Off |
| Normalize             | Off |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FoV Read                 | 208 mm            |
| FoV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8000.0 ms         |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 1                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| CoilShim              | Off   |
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | Off   |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8000.0 ms |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |         |
|-------------------------|---------|
| GLM Statistics          | Off     |
| Ignore Meas. at Start   | 0       |
| Ignore After Transition | 0       |
| Model Transition States | Off     |
| Temp. Highpass Filter   | Off     |
| Threshold               | 4.00    |
| Paradigm Size           | 3       |
| Meas[1]                 | Active  |
| Meas[2]                 | Active  |
| Meas[3]                 | Ignore  |
| Motion Correction       | Off     |
| Spatial Filter          | Off     |
| Measurements            | 1       |
| Delay in TR             | 0.00 ms |
| Multiple Series         | Off     |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                 |   |
|-----------------|---|
| Min. prep scans | 0 |
|-----------------|---|

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| SENSE1 coil combine      | Off        |
| Invert RO/PE polarity    | On         |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

**\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Checkerboard\_AP**TA: 7:48 min Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc.: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 216      |
| Delay in TR  | 0.00 ms  |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 216     |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | Off        |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

**\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Checkerboard\_PA**TA: 48 sec Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 6        |
| Delay in TR  | 0.00 ms  |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 6       |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | On         |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Con1\_AP

TA: 13:56 min Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 400      |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 400     |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | Off        |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |



## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Con1\_PA

TA: 48 sec Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 6        |
| Delay in TR  | 0.00 ms  |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 6       |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | On         |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Con2\_AP

TA: 13:56 min Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 400      |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 400     |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | Off        |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Faces\_AP

TA: 8:36 min Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc.: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 240      |
| Delay in TR  | 0.00 ms  |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 240     |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | Off        |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Faces\_PA

TA: 48 sec Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 6        |
| Delay in TR  | 0.00 ms  |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 6       |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | On         |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |



\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\dMRI\_dir99\_AP

TA: 6:12 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 210 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 210 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 1.5 mm  |
| Base Resolution  | 140     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|            |     |
|------------|-----|
| Raw Filter | Off |
|------------|-----|

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Elliptical Filter     | Off |
| Distortion Correction | Off |
| Normalize             | Off |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 210 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator****Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| CoilShim              | Off   |
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | Off   |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|                          |     |
|--------------------------|-----|
| Resp. Control            | Off |
| Multi-band accel. factor | 4   |

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 99                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | Off                    |
| Trace Weighted Images    | Off                    |
| Tensor                   | Off                    |
| FA Maps                  | Off                    |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3840 us    |
| Refocus pulse duration   | 7680 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| Time-shifted MB RF       | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | Off        |
| PF omits higher k-space  | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Physio recording         | Off        |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\dMRI\_dir99\_PA

TA: 23 sec Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm³ Acc.: None Rel. SNR: 1.00

**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 210 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 210 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 1.5 mm  |
| Base Resolution  | 140     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|            |     |
|------------|-----|
| Raw Filter | Off |
|------------|-----|

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Elliptical Filter     | Off |
| Distortion Correction | Off |
| Normalize             | Off |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 210 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator****Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| CoilShim              | Off   |
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | Off   |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|                          |     |
|--------------------------|-----|
| Resp. Control            | Off |
| Multi-band accel. factor | 4   |

**Diff**

|                          |                     |
|--------------------------|---------------------|
| Diffusion Mode           | Free                |
| Diff. Directions         | 6                   |
| Diffusion Scheme         | Monopolar           |
| Diff. Weightings         | 1                   |
| b-value                  | 0 s/mm <sup>2</sup> |
| Averages                 | 1                   |
| Dynamic Field Correction | Off                 |
| Invert Gray Scale        | Off                 |
| Diff. Weighted Images    | On                  |
| Trace Weighted Images    | Off                 |
| Tensor                   | Off                 |
| FA Maps                  | Off                 |
| ADC Maps                 | Off                 |
| Exponential ADC Maps     | Off                 |
| Calculated Image         | Off                 |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|              |    |
|--------------|----|
| Introduction | On |
|--------------|----|

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 2560 us    |
| Refocus pulse duration   | 5120 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| Time-shifted MB RF       | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | On         |
| PF omits higher k-space  | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Physio recording         | Off        |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\SpinEchoFieldMap\_AP

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

### Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

### Routine

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 69           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 210 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 8000.0 ms    |
| TE                       | 66.00 ms     |
| Multi-band accel. factor | 1            |
| AutoAlign                | Head > Brain |

### Contrast - Common

|                       |                |
|-----------------------|----------------|
| TR                    | 8000.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

### Contrast - Dynamic

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Measurements    | 1        |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

### Resolution - Common

|                  |         |
|------------------|---------|
| FoV Read         | 210 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

### Resolution - Acceleration

|                       |      |
|-----------------------|------|
| Acceleration mode     | None |
| Phase Partial Fourier | Off  |

### Resolution - Filter

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Hamming               | Off |
| Distortion Correction | Off |
| Normalize             | Off |

### Geometry - Common

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 69          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 210 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 8000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 1           |

### Geometry - AutoAlign

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

### Geometry - Saturation

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

### Geometry - Tim Planning Suite

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

### System - Miscellaneous

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

### System - Adjustments

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| CoilShim              | Off   |
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | On    |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8000.0 ms |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |         |
|-------------------------|---------|
| GLM Statistics          | Off     |
| Ignore Meas. at Start   | 0       |
| Ignore After Transition | 0       |
| Model Transition States | Off     |
| Temp. Highpass Filter   | Off     |
| Threshold               | 4.00    |
| Paradigm Size           | 3       |
| Meas[1]                 | Active  |
| Meas[2]                 | Active  |
| Meas[3]                 | Ignore  |
| Motion Correction       | Off     |
| Spatial Filter          | Off     |
| Measurements            | 1       |
| Delay in TR             | 0.00 ms |
| Multiple Series         | Off     |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                 |   |
|-----------------|---|
| Min. prep scans | 0 |
|-----------------|---|

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| SENSE1 coil combine      | Off        |
| Invert RO/PE polarity    | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\SpinEchoFieldMap\_PA

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00

### Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

### Routine

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 69           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 210 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 8000.0 ms    |
| TE                       | 66.00 ms     |
| Multi-band accel. factor | 1            |
| AutoAlign                | Head > Brain |

### Contrast - Common

|                       |                |
|-----------------------|----------------|
| TR                    | 8000.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

### Contrast - Dynamic

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Measurements    | 1        |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

### Resolution - Common

|                  |         |
|------------------|---------|
| FoV Read         | 210 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

### Resolution - Acceleration

|                       |      |
|-----------------------|------|
| Acceleration mode     | None |
| Phase Partial Fourier | Off  |

### Resolution - Filter

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Hamming               | Off |
| Distortion Correction | Off |
| Normalize             | Off |

### Geometry - Common

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 69          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 210 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 8000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 1           |

### Geometry - AutoAlign

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

### Geometry - Saturation

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

### Geometry - Tim Planning Suite

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

### System - Miscellaneous

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

### System - Adjustments

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| CoilShim              | Off   |
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | On    |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8000.0 ms |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |         |
|-------------------------|---------|
| GLM Statistics          | Off     |
| Ignore Meas. at Start   | 0       |
| Ignore After Transition | 0       |
| Model Transition States | Off     |
| Temp. Highpass Filter   | Off     |
| Threshold               | 4.00    |
| Paradigm Size           | 3       |
| Meas[1]                 | Active  |
| Meas[2]                 | Active  |
| Meas[3]                 | Ignore  |
| Motion Correction       | Off     |
| Spatial Filter          | Off     |
| Measurements            | 1       |
| Delay in TR             | 0.00 ms |
| Multiple Series         | Off     |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                 |   |
|-----------------|---|
| Min. prep scans | 0 |
|-----------------|---|

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| SENSE1 coil combine      | Off        |
| Invert RO/PE polarity    | On         |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |



\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\pd+t2\_tse\_tra

TA: 2:11 min Coil Selection: Auto Voxel Size: 0.3×0.3×4.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00

### Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

### Resolution - Acceleration

|                         |            |
|-------------------------|------------|
| Acceleration mode       | GRAPPA     |
| Reference Scans         | Integrated |
| Acceleration Factor PE  | 2          |
| Reference Lines PE      | 30         |
| Advanced Reconstruction | Off        |
| Phase Partial Fourier   | Off        |

### Resolution - Filter

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | On      |
| Distortion Correction | 2D      |
| Normalize             | Prescan |
| Image Filter          | Off     |

### Routine

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Slices              | 30                |
| Distance Factor     | 25 %              |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | R >> L            |
| Phase Oversampling  | 0 %               |
| FoV Read            | 220 mm            |
| FoV Phase           | 100.0 %           |
| Slice Thickness     | 4.0 mm            |
| TR                  | 3600.0 ms         |
| TE 1                | 9.40 ms           |
| TE 2                | 94.00 ms          |
| Averages            | 1                 |
| Concatenations      | 1                 |
| AutoAlign           | Head > Brain      |

### Geometry - Common

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Slices              | 30                |
| Distance Factor     | 25 %              |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | R >> L            |
| Phase Oversampling  | 0 %               |
| FoV Read            | 220 mm            |
| FoV Phase           | 100.0 %           |
| Slice Thickness     | 4.0 mm            |
| TR                  | 3600.0 ms         |
| Multi-Slice Mode    | Interleaved       |
| Series              | Interleaved       |
| Concatenations      | 1                 |

### Contrast - Common

|                    |           |
|--------------------|-----------|
| TR                 | 3600.0 ms |
| TE 1               | 9.40 ms   |
| TE 2               | 94.00 ms  |
| MTC                | Off       |
| Magn. Preparation  | None      |
| Flip Angle         | 160 deg   |
| Fat-Water Contrast | Standard  |
| Dark Blood         | Off       |
| Contrasts          | 2         |
| Wrap-up Magn.      | None      |
| Reconstruction     | Magnitude |

### Geometry - AutoAlign

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | R >> L            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 A2.1 F2.7    |
| R                   | 0.0 mm            |
| A                   | 2.1 mm            |
| F                   | 2.7 mm            |
| Initial Orientation | T > C             |
| T > C               | -5.90             |
| > S                 | 0.00              |
| Initial Rotation    | 89.87 deg         |

### Contrast - Dynamic

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |

### Geometry - Navigator

### Geometry - Saturation

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

### Resolution - Common

|                  |           |
|------------------|-----------|
| FoV Read         | 220 mm    |
| FoV Phase        | 100.0 %   |
| Slice Thickness  | 4.0 mm    |
| Base Resolution  | 320       |
| Phase Resolution | 100 %     |
| Trajectory       | Cartesian |
| Interpolation    | On        |

### Geometry - Tim Planning Suite

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

### System - Miscellaneous

|                |                   |
|----------------|-------------------|
| Coil Selection | ACS All but spine |
| MSMA           | S - C - T         |
| Sagittal       | R >> L            |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Adaptive Combine |
| Matrix Optimization | Off              |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Tune up  |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|              |          |
|--------------|----------|
| B1 Shim      | TrueForm |
| LR Balancing | Off      |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |           |
|-----------------|-----------|
| 1st Signal/Mode | None      |
| TR              | 3600.0 ms |
| Concatenations  | 1         |

**Physio - Cardiac**

|                    |           |
|--------------------|-----------|
| Fat-Water Contrast | Standard  |
| Magn. Preparation  | None      |
| Dark Blood         | Off       |
| FoV Read           | 220 mm    |
| FoV Phase          | 100.0 %   |
| Phase Resolution   | 100 %     |
| Trajectory         | Cartesian |
| Dynamic Mode       | Standard  |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 1   |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |      |
|----------------------|------|
| Magn. Preparation    | None |
| Save Original Images | On   |

**Inline - Cardiac**

|           |           |
|-----------|-----------|
| Contrasts | 2         |
| TE 1      | 9.40 ms   |
| TE 2      | 94.00 ms  |
| TR        | 3600.0 ms |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Sequence - Part 1**

|                       |              |
|-----------------------|--------------|
| Sequence Name         | tse          |
| Dimension             | 2D           |
| RF Pulse Type         | Normal       |
| Gradient Mode         | Normal       |
| Flow Compensation     | None         |
| Bandwidth             | 252 Hz/Px    |
| Echo Spacing          | 9.36 ms      |
| Free Echo Spacing     | Off          |
| Define                | Turbo Factor |
| Turbo Factor          | 5            |
| Echo Trains per Slice | 35           |

**Sequence - Part 2**

|                          |           |
|--------------------------|-----------|
| Introduction             | On        |
| Phase Correction         | Automatic |
| Compensate T2 Decay      | Off       |
| Hyperecho                | Off       |
| WARP                     | Off       |
| Red. EC Sensitivity      | Off       |
| Acoustic noise reduction | Off       |
| Reduce Motion Sens.      | On        |

**Sequence - Assistant**

|                |            |
|----------------|------------|
| SAR Assistant  | Flip Angle |
| Min Flip Angle | 130 deg    |
| Allowed Delay  | 60 s       |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\t2\_tse\_dark-fluid\_tra

TA: 4:14 min Coil Selection: Auto Voxel Size: 0.7×0.7×4.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | On                 |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Slices              | 30                |
| Distance Factor     | 25 %              |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | R >> L            |
| Phase Oversampling  | 0 %               |
| FoV Read            | 220 mm            |
| FoV Phase           | 96.9 %            |
| Slice Thickness     | 4.0 mm            |
| TR                  | 9000.0 ms         |
| TE                  | 83.00 ms          |
| Averages            | 1                 |
| Concatenations      | 2                 |
| AutoAlign           | Head > Brain      |

**Contrast - Common**

|                      |                |
|----------------------|----------------|
| TR                   | 9000.0 ms      |
| TE                   | 83.00 ms       |
| TD                   | 0.00 ms        |
| MTC                  | Off            |
| Magn. Preparation    | Slice-sel. IR  |
| TI                   | 2500 ms        |
| Freeze Suppr. Tissue | On             |
| Flip Angle           | 150 deg        |
| Fat-Water Contrast   | Fat Saturation |
| Fat Saturation       | Strong         |
| Dark Blood           | Off            |
| Contrasts            | 1              |
| Wrap-up Magn.        | None           |
| Reconstruction       | Magnitude      |

**Contrast - Dynamic**

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |

**Resolution - Common**

|                  |           |
|------------------|-----------|
| FoV Read         | 220 mm    |
| FoV Phase        | 96.9 %    |
| Slice Thickness  | 4.0 mm    |
| Base Resolution  | 320       |
| Phase Resolution | 70 %      |
| Trajectory       | Cartesian |
| Interpolation    | Off       |

**Resolution - Acceleration**

|                         |      |
|-------------------------|------|
| Acceleration mode       | None |
| Advanced Reconstruction | Off  |
| Phase Partial Fourier   | Off  |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | On      |
| Distortion Correction | 2D      |
| Normalize             | Prescan |
| Image Filter          | Off     |

**Geometry - Common**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Slices              | 30                |
| Distance Factor     | 25 %              |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | R >> L            |
| Phase Oversampling  | 0 %               |
| FoV Read            | 220 mm            |
| FoV Phase           | 96.9 %            |
| Slice Thickness     | 4.0 mm            |
| TR                  | 9000.0 ms         |
| Multi-Slice Mode    | Interleaved       |
| Series              | Interleaved       |
| Concatenations      | 2                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | R >> L            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 A2.1 F2.7    |
| R                   | 0.0 mm            |
| A                   | 2.1 mm            |
| F                   | 2.7 mm            |
| Initial Orientation | T > C             |
| T > C               | -5.90             |
| > S                 | 0.00              |
| Initial Rotation    | 89.87 deg         |

**Geometry - Navigator****Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                  |                   |
|------------------|-------------------|
| Coil Selection   | ACS All but spine |
| MSMA             | S - C - T         |
| Sagittal         | R >> L            |
| Coronal          | A >> P            |
| Transversal      | F >> H            |
| Coil Combination | Adaptive Combine  |

**System - Miscellaneous**

|                     |      |
|---------------------|------|
| Matrix Optimization | Off  |
| Coil Focus          | Flat |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 A2.1 F2.7 mm |
| Orientation | T > C-5.9         |
| Rotation    | 89.87 deg         |
| R >> L      | 214 mm            |
| A >> P      | 220 mm            |
| F >> H      | 149 mm            |
| Reset       | Off               |

**System - pTx**

|              |          |
|--------------|----------|
| B1 Shim      | TrueForm |
| LR Balancing | Off      |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |           |
|-----------------|-----------|
| 1st Signal/Mode | None      |
| TR              | 9000.0 ms |
| Concatenations  | 2         |

**Physio - Cardiac**

|                    |                |
|--------------------|----------------|
| Fat-Water Contrast | Fat Saturation |
| Magn. Preparation  | Slice-sel. IR  |
| TI                 | 2500 ms        |
| Dark Blood         | Off            |
| FoV Read           | 220 mm         |
| FoV Phase          | 96.9 %         |
| Phase Resolution   | 70 %           |
| Trajectory         | Cartesian      |
| Dynamic Mode       | Standard       |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 2   |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |               |
|----------------------|---------------|
| Magn. Preparation    | Slice-sel. IR |
| Save Original Images | On            |
| Contrasts            | 1             |
| TE                   | 83.00 ms      |

**Inline - Cardiac**

|    |           |
|----|-----------|
| TR | 9000.0 ms |
|----|-----------|

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Sequence - Part 1**

|                       |              |
|-----------------------|--------------|
| Sequence Name         | tir          |
| Dimension             | 2D           |
| RF Pulse Type         | Normal       |
| Gradient Mode         | Performance  |
| Flow Compensation     | None         |
| Bandwidth             | 289 Hz/Px    |
| Echo Spacing          | 8.28 ms      |
| Free Echo Spacing     | Off          |
| Define                | Turbo Factor |
| Turbo Factor          | 17           |
| Echo Trains per Slice | 13           |

**Sequence - Part 2**

|                          |           |
|--------------------------|-----------|
| Introduction             | On        |
| Phase Correction         | Automatic |
| Compensate T2 Decay      | Off       |
| Hyperecho                | Off       |
| WARP                     | Off       |
| Red. EC Sensitivity      | Off       |
| Acoustic noise reduction | Off       |
| Reduce Motion Sens.      | On        |

**Sequence - Assistant**

|               |      |
|---------------|------|
| SAR Assistant | Off  |
| Allowed Delay | 60 s |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\resolve\_4scan\_trace\_trace\_p2\_192

TA: 1:51 min Coil Selection: Auto Voxel Size: 1.1×1.1×4.0 mm<sup>3</sup> Acc.: 2 Rel. SNR: 1.00

### Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

### Routine

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Slices              | 30                |
| Distance Factor     | 25 %              |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | A >> P            |
| Phase Oversampling  | 0 %               |
| FoV Read            | 220 mm            |
| FoV Phase           | 100.0 %           |
| Slice Thickness     | 4.0 mm            |
| TR                  | 4040.0 ms         |
| TE 1                | 55 ms             |
| TE 2                | 93 ms             |
| Concatenations      | 1                 |
| AutoAlign           | Head > Brain      |

### Contrast - Common

|                    |                |
|--------------------|----------------|
| TR                 | 4040.0 ms      |
| TE 1               | 55 ms          |
| TE 2               | 93 ms          |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 180 deg        |
| Fat-Water Contrast | Fat Saturation |
| Fat Saturation     | Strong         |
| Contrasts          | 2              |
| Reconstruction     | Magnitude      |

### Contrast - Dynamic

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 1        |

### Resolution - Common

|                  |         |
|------------------|---------|
| FoV Read         | 220 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 4.0 mm  |
| Base Resolution  | 192     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

### Resolution - Acceleration

|                        |              |
|------------------------|--------------|
| Accel. Mode            | GRAPPA       |
| Reference Scans        | EPI/Separate |
| Acceleration Factor PE | 2            |

### Resolution - Acceleration

|                         |     |
|-------------------------|-----|
| Reference Lines PE      | 96  |
| Phase Partial Fourier   | Off |
| Readout Partial Fourier | 5/8 |
| Readout Segments        | 7   |

### Resolution - Filter

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Distortion Correction | Off     |
| Normalize             | Prescan |
| Noise Masking         | Off     |

### Geometry - Common

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Slices              | 30                |
| Distance Factor     | 25 %              |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | A >> P            |
| Phase Oversampling  | 0 %               |
| FoV Read            | 220 mm            |
| FoV Phase           | 100.0 %           |
| Slice Thickness     | 4.0 mm            |
| TR                  | 4040.0 ms         |
| Multi-Slice Mode    | Interleaved       |
| Series              | Interleaved       |
| Concatenations      | 1                 |

### Geometry - AutoAlign

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 A2.1 F2.7 mm |
| Orientation         | T > C-5.9         |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 A2.1 F2.7    |
| R                   | 0.0 mm            |
| A                   | 2.1 mm            |
| F                   | 2.7 mm            |
| Initial Orientation | T > C             |
| T > C               | -5.90             |
| > S                 | 0.00              |
| Initial Rotation    | -0.13 deg         |

### Geometry - Saturation

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

### Geometry - Tim Planning Suite

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 3 mm |
| Table Position    | F    |
| Inline Composing  | Off  |

### System - Miscellaneous

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Adaptive Combine  |
| Matrix Optimization | Off               |

**System - Miscellaneous**

|            |      |
|------------|------|
| Coil Focus | Flat |
|------------|------|

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 A2.1 F2.7 mm |
| Orientation | T > C-5.9         |
| Rotation    | -0.13 deg         |
| A >> P      | 220 mm            |
| R >> L      | 220 mm            |
| F >> H      | 149 mm            |
| Reset       | Off               |

**System - pTx**

|         |          |
|---------|----------|
| B1 Shim | TrueForm |
|---------|----------|

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |           |
|-----------------|-----------|
| 1st Signal/Mode | None      |
| TR              | 4040.0 ms |
| Concatenations  | 1         |

**Diff**

|                       |                        |
|-----------------------|------------------------|
| Diffusion Mode        | 4-Scan Trace           |
| Diff. Directions      | 4                      |
| Diffusion Scheme      | Monopolar              |
| Diff. Weightings      | 2                      |
| b-value 1             | 0 s/mm <sup>2</sup>    |
| b-value 2             | 1000 s/mm <sup>2</sup> |
| Averages 1            | 1                      |
| Averages 2            | 1                      |
| Invert Gray Scale     | Off                    |
| Diff. Weighted Images | Off                    |
| Trace Weighted Images | On                     |
| Tensor                | Off                    |
| FA Maps               | Off                    |
| ADC Maps              | On                     |
| Exponential ADC Maps  | Off                    |
| b-value >=            | 0 s/mm <sup>2</sup>    |
| ADC Noise Threshold   | 100                    |
| Noise Masking         | Off                    |
| Calculated Image      | Off                    |

**Sequence - Part 1**

|               |           |
|---------------|-----------|
| Sequence Name | resolve   |
| Dimension     | 2D        |
| RF Pulse Type | Normal    |
| Gradient Mode | Fast      |
| Bandwidth     | 766 Hz/Px |
| Echo Spacing  | 0.32 ms   |

**Sequence - Part 1**

|              |         |
|--------------|---------|
| Optimization | Min. TE |
| EPI Factor   | 96      |

**Sequence - Part 2**

|                    |    |
|--------------------|----|
| Introduction       | On |
| Reacquisition Mode | On |

**Sequence - Assistant**

|               |         |
|---------------|---------|
| SAR Assistant | Off     |
| Optimization  | Min. TE |

## \\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\Resting\_AP

TA: 7:48 min Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm³ Acc:: 2 Rel. SNR: 1.00

**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 216      |
| Delay in TR  | 0.00 ms  |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 216     |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | Off        |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |



**\\Research\Investigators - validated on FIT\Harnett\ATE\_Study\_6\_21\_2023\Resting\_PA**
TA: 48 sec Coil Selection: Auto Voxel Size: 2.4×2.4×2.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | On      |
| Elliptical Filter     | Off     |
| Hamming               | Off     |
| Distortion Correction | 2D      |
| Normalize             | Prescan |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 72           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FoV Read                 | 208 mm       |
| FoV Phase                | 100.0 %      |
| Slice Thickness          | 2.0 mm       |
| TR                       | 2000.0 ms    |
| TE 1                     | 13.00 ms     |
| TE 2                     | 30.22 ms     |
| TE 3                     | 47.44 ms     |
| TE 4                     | 64.66 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 72          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FoV Read                 | 208 mm      |
| FoV Phase                | 100.0 %     |
| Slice Thickness          | 2.0 mm      |
| TR                       | 2000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 2000.0 ms      |
| TE 1               | 13.00 ms       |
| TE 2               | 30.22 ms       |
| TE 3               | 47.44 ms       |
| TE 4               | 64.66 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magn./Phase    |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 6        |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 208 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                  |                  |
|------------------|------------------|
| Coil Selection   | Auto Coil Select |
| MSMA             | S - C - T        |
| Sagittal         | R >> L           |
| Coronal          | A >> P           |
| Transversal      | F >> H           |
| Coil Combination | Sum of Squares   |

**System - Miscellaneous**

|                     |             |
|---------------------|-------------|
| Matrix Optimization | Performance |
| Coil Focus          | Flat        |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 208 mm     |
| R >> L      | 208 mm     |
| F >> H      | 144 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 2000.0 ms |
| Multi-band accel. factor | 4         |

**BOLD**

|                         |        |
|-------------------------|--------|
| GLM Statistics          | Off    |
| Ignore Meas. at Start   | 0      |
| Ignore After Transition | 0      |
| Model Transition States | On     |
| Temp. Highpass Filter   | On     |
| Threshold               | 4.00   |
| Paradigm Size           | 20     |
| Meas[1]                 | Active |
| Meas[2]                 | Active |
| Meas[3]                 | Ignore |
| Meas[4]                 | Ignore |
| Meas[5]                 | Ignore |
| Meas[6]                 | Ignore |
| Meas[7]                 | Ignore |
| Meas[8]                 | Ignore |
| Meas[9]                 | Ignore |
| Meas[10]                | Ignore |
| Meas[11]                | Ignore |
| Meas[12]                | Ignore |
| Meas[13]                | Ignore |
| Meas[14]                | Ignore |
| Meas[15]                | Ignore |
| Meas[16]                | Ignore |
| Meas[17]                | Ignore |
| Meas[18]                | Ignore |
| Meas[19]                | Ignore |

**BOLD**

|                   |         |
|-------------------|---------|
| Meas[20]          | Ignore  |
| Motion Correction | Off     |
| Spatial Filter    | Off     |
| Measurements      | 6       |
| Delay in TR       | 0.00 ms |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.52 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |            |
|--------------------------|------------|
| Excite pulse duration    | 3820 us    |
| Min. prep scans          | 0          |
| Min. prep scans SB       | 0          |
| Inter-TE delay           | 0 us       |
| Single-band images       | On         |
| MB LeakBlock kernel      | On         |
| MB dual kernel           | Off        |
| MB RF phase scramble     | On         |
| Opt. MB RF pulse BW      | Off        |
| SENSE1 coil combine      | On         |
| Invert RO/PE polarity    | On         |
| Force Maxwell corr.      | Off        |
| Disable freq. update     | Off        |
| Suppress 16-bit DICOM    | On         |
| Force equal slice timing | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Fat saturation FA        | 110.00 deg |
| Fat sat. offset          | 0.00 Hz    |
| Sinc exc. pulse BWTP     | 5.20       |
| Physio recording         | Off        |
| Triggering scheme        | Standard   |

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\ABCD\_T2w\_SPC\_vNav\_s  
etter

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm<sup>3</sup> Rel. SNR: 1.00

### Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

### Routine

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 50 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 32                 |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FoV Read            | 256 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 8.0 mm             |
| TR                  | 13.0 ms            |
| TE                  | 6.10 ms            |
| Averages            | 1                  |
| Concatenations      | 1                  |
| AutoAlign           | ---                |
| Coil Elements       | BC                 |

### Contrast - Common

|                    |                  |
|--------------------|------------------|
| TR                 | 13.0 ms          |
| TE                 | 6.10 ms          |
| MTC                | Off              |
| Flip Angle         | 2 deg            |
| Fat-Water Contrast | Water Excitation |
| Reconstruction     | Magnitude        |

### Contrast - Dynamic

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |

### Resolution - Common

|                  |         |
|------------------|---------|
| FoV Read         | 256 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 8.0 mm  |
| Base Resolution  | 32      |
| Phase Resolution | 100 %   |
| Slice Resolution | 100 %   |
| Interpolation    | Off     |

### Resolution - Acceleration

|                       |     |
|-----------------------|-----|
| Phase Partial Fourier | Off |
| Slice Partial Fourier | 6/8 |

### Resolution - Filter

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Hamming               | Off |
| Distortion Correction | Off |
| Normalize             | Off |

### Geometry - Common

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 50 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 32                 |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FoV Read            | 256 mm             |
| FoV Phase           | 100.0 %            |
| Slice Thickness     | 8.0 mm             |
| TR                  | 13.0 ms            |
| Multi-Slice Mode    | Interleaved        |
| Series              | Interleaved        |
| Concatenations      | 1                  |

### Geometry - AutoAlign

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| AutoAlign           | ---                |
| Initial Position    | L0.0 A20.0 H0.0    |
| L                   | 0.0 mm             |
| A                   | 20.0 mm            |
| F                   | 0.0 mm             |
| Initial Orientation | Sagittal           |
| Initial Rotation    | 0.00 deg           |

### Geometry - Saturation

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

### Geometry - Tim Planning Suite

|                   |       |
|-------------------|-------|
| Set-n-Go Protocol | Off   |
| Table Position    | 23 mm |
| Table Position    | F     |
| Inline Composing  | Off   |

### System - Miscellaneous

|                     |                |
|---------------------|----------------|
| Coil Selection      | Manual         |
| MSMA                | S - C - T      |
| Sagittal            | R >> L         |
| Coronal             | A >> P         |
| Transversal         | F >> H         |
| Coil Combination    | Sum of Squares |
| Matrix Optimization | Off            |

### System - Adjustments

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
| B0 Shim             | Standard |
| B1 Shim             | TrueForm |

**System - Adjustments**

|                       |       |
|-----------------------|-------|
| Adjustment Tolerance  | Auto  |
| Adjust with Body Coil | Off   |
| Confirm Frequency     | Never |
| Assume Silicone       | Off   |

**System - Adjust Volume**

|             |                    |
|-------------|--------------------|
| Position    | L0.0 A20.0 H0.0 mm |
| Orientation | Sagittal           |
| Rotation    | 0.00 deg           |
| A >> P      | 256 mm             |
| F >> H      | 256 mm             |
| R >> L      | 256 mm             |
| Reset       | Off                |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Non-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Correction Factor   | 1.00           |
| Image Scaling       | 8.000          |

**Physio - Signal**

|                 |         |
|-----------------|---------|
| 1st Signal/Mode | None    |
| TR              | 13.0 ms |
| Concatenations  | 1       |

**Sequence - Part 1**

|                   |            |
|-------------------|------------|
| Sequence Name     | ABCD       |
| Dimension         | 3D         |
| Excitation        | Non-sel.   |
| RF Pulse Type     | Normal     |
| Gradient Mode     | Fast       |
| Bandwidth         | 4882 Hz/Px |
| Echo Spacing      | 0.27 ms    |
| Free Echo Spacing | Off        |
| EPI Factor        | 32         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | On  |

**Sequence - Special**

|                   |          |
|-------------------|----------|
| Protocol filename | T2-SPACE |
|-------------------|----------|

\\Research\\Investigators - validated on FIT\\Harnett\\ATE\_Study\_6\_21\_2023\\ABCD\_T2w\_SPC\_vNav

TA: 7:36 min Coil Selection: Auto Voxel Size: 1.0×1.0×0.8 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                     |                     |
|---------------------|---------------------|
| Slab Group          | 1                   |
| Slabs               | 1                   |
| Position            | L0.0 A20.0 F30.0 mm |
| Orientation         | Sagittal            |
| Phase Encoding Dir. | A >> P              |
| Slices per Slab     | 208                 |
| Phase Oversampling  | 0 %                 |
| Slice Oversampling  | 0.0 %               |
| FoV Read            | 256 mm              |
| FoV Phase           | 100.0 %             |
| Slice Thickness     | 0.80 mm             |
| TR                  | 3200.0 ms           |
| TE                  | 562.00 ms           |
| Averages            | 1.0                 |
| Concatenations      | 1                   |
| AutoAlign           | ---                 |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| TR                 | 3200.0 ms |
| TE                 | 562.00 ms |
| MTC                | Off       |
| Magn. Preparation  | None      |
| Flip Angle Mode    | T2 Var    |
| Fat-Water Contrast | Standard  |
| Dark Blood         | Off       |
| Blood Suppression  | Off       |
| Wrap-up Magn.      | None      |
| Reconstruction     | Magnitude |

**Contrast - Dynamic**

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |
| Reordering      | Linear           |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FoV Read         | 256 mm  |
| FoV Phase        | 100.0 % |
| Slice Thickness  | 0.80 mm |
| Base Resolution  | 256     |
| Phase Resolution | 100 %   |
| Slice Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                        |            |
|------------------------|------------|
| Acceleration mode      | GRAPPA     |
| Total Factor           | 2          |
| Reference Scans        | Integrated |
| Acceleration Factor PE | 2          |
| Reference Lines PE     | 32         |
| Acceleration Factor 3D | 1          |
| Phase Partial Fourier  | Allowed    |
| Slice Partial Fourier  | Off        |
| Elliptical Scanning    | On         |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | Off     |
| Distortion Correction | Off     |
| Normalize             | Prescan |
| Image Filter          | Off     |

**Geometry - Common**

|                     |                     |
|---------------------|---------------------|
| Slab Group          | 1                   |
| Slabs               | 1                   |
| Position            | L0.0 A20.0 F30.0 mm |
| Orientation         | Sagittal            |
| Phase Encoding Dir. | A >> P              |
| Slices per Slab     | 208                 |
| Phase Oversampling  | 0 %                 |
| Slice Oversampling  | 0.0 %               |
| FoV Read            | 256 mm              |
| FoV Phase           | 100.0 %             |
| Slice Thickness     | 0.80 mm             |
| TR                  | 3200.0 ms           |
| Concatenations      | 1                   |

**Geometry - AutoAlign**

|                     |                     |
|---------------------|---------------------|
| Slab Group          | 1                   |
| Position            | L0.0 A20.0 F30.0 mm |
| Orientation         | Sagittal            |
| Phase Encoding Dir. | A >> P              |
| AutoAlign           | ---                 |
| Initial Position    | L0.0 A20.0 F30.0    |
| R                   | 0.0 mm              |
| A                   | 20.0 mm             |
| F                   | 30.0 mm             |
| Initial Orientation | Sagittal            |
| Initial Rotation    | 0.00 deg            |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |       |
|-------------------|-------|
| Set-n-Go Protocol | Off   |
| Table Position    | 30 mm |
| Table Position    | F     |
| Inline Composing  | Off   |

**System - Miscellaneous**

|                |                  |
|----------------|------------------|
| Coil Selection | Auto Coil Select |
| MSMA           | S - C - T        |
| Sagittal       | R >> L           |
| Coronal        | A >> P           |
| Transversal    | F >> H           |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Combination    | Adaptive Combine |
| Matrix Optimization | Off              |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| CoilShim              | Off      |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                     |
|-------------|---------------------|
| Position    | L0.0 A20.0 F30.0 mm |
| Orientation | Sagittal            |
| Rotation    | 0.00 deg            |
| A >> P      | 256 mm              |
| F >> H      | 256 mm              |
| R >> L      | 167 mm              |
| Reset       | Off                 |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Non-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253468 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |           |
|-----------------|-----------|
| 1st Signal/Mode | None      |
| Trigger Delay   | 0 ms      |
| TR              | 3200.0 ms |
| Concatenations  | 1         |

**Physio - Cardiac**

|                    |          |
|--------------------|----------|
| Fat-Water Contrast | Standard |
| Magn. Preparation  | None     |
| Dark Blood         | Off      |
| FoV Read           | 256 mm   |
| FoV Phase          | 100.0 %  |
| Phase Resolution   | 100 %    |
| Dynamic Mode       | Standard |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - Cardiac**

|                      |           |
|----------------------|-----------|
| Magn. Preparation    | None      |
| Save Original Images | On        |
| TE                   | 562.00 ms |
| TR                   | 3200.0 ms |

**Inline - MIP**

|         |     |
|---------|-----|
| MIP Sag | Off |
| MIP Cor | Off |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Sequence - Part 1**

|                     |           |
|---------------------|-----------|
| Sequence Name       | spc       |
| Dimension           | 3D        |
| Excitation          | Non-sel.  |
| RF Pulse Type       | Fast      |
| Gradient Mode       | Fast      |
| Flow Compensation   | None      |
| Reordering          | Linear    |
| Bandwidth           | 241 Hz/Px |
| Echo Spacing        | 6.54 ms   |
| Turbo Factor        | 200       |
| Echo Train Duration | 1210 ms   |

**Sequence - Part 2**

|              |    |
|--------------|----|
| Introduction | On |
|--------------|----|

**Sequence - Special**

|                   |                |
|-------------------|----------------|
| Include Nav.      | On             |
| Apply moco to     | parent and nav |
| Remeasure         | 18 TRs         |
| Reacq. threshold  | 0.50           |
| Feedback Delay    | 80 ms          |
| Moco ref. image   | Use Temp Ref   |
| K-space streaming | None           |
| ABCD navigator    | On             |
| Apply freq to     | parent and nav |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
| Allowed Delay | 0 s |